Linear Models Lecture 5: Finite Sample Properties

Robert Gulotty

University of Chicago

April 6, 2026

Why use $(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'Y$ in my work?

- Motive 1: best approximation to the conditional expectation function (CEF).
  - While the conditional mean $E(Y|\boldsymbol{X})$ is the best predictor of Y, its form is typically unknown.
  - The linear model is an approximation to the conditional mean that has the lowest mean squared error among linear predictors.
- Motive 2: Sampling Properties.
  - When the underlying CEF is linear and the residual variance is constant, OLS is optimal.
  - OLS is the Best Linear Unbiased Estimator, here meaning the minimum variance (most efficient) linear estimator for the parameter vector $\boldsymbol{\beta}$.

Why Study Finite Sample Properties?

- Finite sample results hold *exactly* for any sample size n—they do not require $n \to \infty$.
- The finite sample framework answers four questions:
  1. Is $\hat{\beta}$ centered on β? $\to$ **Unbiasedness.**
  2. How much does $\hat{\beta}$ vary across samples? $\to$ **Variance.**
  3. Can we squeeze more precision out of the data?
     - If $\text{var}[\boldsymbol{e}|\boldsymbol{X}] = \sigma^2 \boldsymbol{I}$? $\to$ **Gauss-Markov**
     - If $\text{var}[\boldsymbol{e}|\boldsymbol{X}] \neq \sigma^2 \boldsymbol{I}$? $\to$ **GLS.**
- Efficiency matters especially in political science: treatments are often weak and effective sample sizes are small, so every gain in precision counts.
- These results provide the theoretical foundation for standard errors, confidence intervals, and hypothesis tests.

Finite Sample Assumptions of OLS estimator

- Assumption 1: The random variables $\{(Y_1,\ \boldsymbol{x}_1),\ (Y_2,\ \boldsymbol{x}_2), \dots, (Y_n,\ \boldsymbol{x}_n)\}$ are independent and identically distributed.
- Assumption 2: $Y_i = \boldsymbol{x}_i' \beta + e_i$, where $\mathbb{E}(e_i | \boldsymbol{x}_i) = 0$.
- Assumption 3: $\mathbb{E}(\boldsymbol{x}_i \boldsymbol{x}_i') > 0$ (with probability 1).
- Assumption 4*: $\mathbb{E}[e_i^2 | \boldsymbol{x}_i] = \sigma^2(\boldsymbol{x}_i) = \sigma^2$.

When Are These Assumptions Reasonable?

- **A1 (i.i.d.):** Holds well for cross-sectional surveys with random sampling. Fails with time-series data (serial correlation), clustered data (students within schools), or panel data.
- **A2 ($\mathbb{E}[e|\boldsymbol{X}] = 0$):** Requires that no omitted variable is correlated with $\boldsymbol{X}$. Fails when there is selection bias, simultaneity, or measurement error in $\boldsymbol{X}$. This is the assumption most contested in applied work.
- **A3 (rank condition):** Fails with perfect multicollinearity (e.g. including a dummy for every category plus an intercept). Nearly violated when regressors are highly collinear.
- **A4* (homoskedasticity):** Rarely holds exactly. Variance of earnings typically grows with education; variance of GDP growth differs across countries. Violation does *not* bias $\hat{\beta}$, but makes OLS inefficient and standard errors wrong.

Goals for Sampling Properties.

- **unbiased**: The (conditional) expectation of the estimator $\hat{\beta}$ of β is equal to the paramater. $\mathbb{E}[\hat{\beta}] = \beta$
- **efficient**: The estimator $\hat{\beta}$ of β has a lower variance matrix than other estimators.
- The latter efficiency result is called the Gauss Markov Theorem and depends on Assumption 4*.

Unbiasedness

- We will show that $\mathbb{E}[\hat{\beta}|\boldsymbol{X}] = \beta$ using three methods: summation, matrix, and decomposition.
- Each relies heavily on the conditioning theorem:

$$\mathbb{E}[g(\boldsymbol{X})Y|\boldsymbol{X}] = g(\boldsymbol{X})\mathbb{E}[Y|\boldsymbol{X}]$$

and

$$\mathbb{E}[g(\boldsymbol{X})Y] = \mathbb{E}[g(\boldsymbol{X})\mathbb{E}[Y|\boldsymbol{X}]]$$

Unbiasedness of the OLS slope estimator: Version 1

- Assume $\boldsymbol{x}_i$ is a $k \times 1$ vector representing observation i
- $\mathbb{E}[Y_i|\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n] = \mathbb{E}[Y_i|\boldsymbol{x}_i]$ because of independence of observations across i.

Unbiasedness (shown in summation operation)

$$
\begin{aligned}
\mathbb{E}[\hat{\beta}|\boldsymbol{x}_1, \dots \boldsymbol{x}_n] &= \mathbb{E}\left[(\sum_{i=1}^{n} \boldsymbol{x}_i \boldsymbol{x}_i')^{-1} (\sum_{i=1}^{n} \boldsymbol{x}_i Y_i) \;\middle|\; \boldsymbol{x}_1, \dots \boldsymbol{x}_n \right] \\
&= (\sum_{i=1}^{n} \boldsymbol{x}_i \boldsymbol{x}_i')^{-1} \mathbb{E}[(\sum_{i=1}^{n} \boldsymbol{x}_i Y_i)|\boldsymbol{x}_1, \dots \boldsymbol{x}_n] && \text{(Cond. Theorem)} \\
&= (\sum_{i=1}^{n} \boldsymbol{x}_i \boldsymbol{x}_i')^{-1} \sum_{i=1}^{n} \mathbb{E}[\boldsymbol{x}_i Y_i|\boldsymbol{x}_1, \dots \boldsymbol{x}_n] && \text{(Linearity)} \\
&= (\sum_{i=1}^{n} \boldsymbol{x}_i \boldsymbol{x}_i')^{-1} \sum_{i=1}^{n} \boldsymbol{x}_i \mathbb{E}[Y_i|\boldsymbol{x}_i] && \text{(Cond. Theorem + indep.)} \\
&= (\sum_{i=1}^{n} \boldsymbol{x}_i \boldsymbol{x}_i')^{-1} \sum_{i=1}^{n} \boldsymbol{x}_i \boldsymbol{x}_i' \beta && \text{(Linear CEF)} \\
&= \beta && \text{(Inverse)}
\end{aligned}
$$

Unbiasedness of the OLS slope estimator: Version 2

Using Matrix notation, $\mathbb{E}[\boldsymbol{y}|\boldsymbol{X}] = \boldsymbol{X}\beta$, $\mathbb{E}[\boldsymbol{e}|\boldsymbol{X}] = 0$ (by assumption 2)

$$\begin{aligned}
\mathbb{E}[\hat{\beta}|\boldsymbol{X}] &= \mathbb{E}[(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}|\boldsymbol{X}] \\
&= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\mathbb{E}[\boldsymbol{y}|\boldsymbol{X}] && \text{(Conditioning Theorem)} \\
&= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{X}\beta && \text{(Independence)} \\
&= \beta && \text{(Inverse)}
\end{aligned}$$

Unbiasedness of the OLS slope estimator: Version 3

Decomposition, noting $\boldsymbol{y} = \boldsymbol{X}\beta + \boldsymbol{e}$

$$
\begin{aligned}
\hat{\beta} &= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y} \\
&= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'(\boldsymbol{X}\beta + \boldsymbol{e}) && \text{(plug in)} \\
&= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{X}\beta + (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{e} && \text{(distribute)} \\
&= \beta + (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{e} && \text{(Inverse)}
\end{aligned}
$$

So now we can just check that $\mathbb{E}[\hat{\beta} - \beta|\boldsymbol{X}] = 0$ by applying assumption 2.

$$
\mathbb{E}[\hat{\beta} - \beta|\boldsymbol{X}] = \mathbb{E}[(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{e}|\boldsymbol{X}] = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\mathbb{E}[\boldsymbol{e}|\boldsymbol{X}] = 0
$$

Variance of Least Squares Estimator

We can write the variance of the OLS estimator in terms of

$$\boldsymbol{D} = \begin{pmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \sigma_n^2 \end{pmatrix}$$

Where σ_i^2 gives us the variation in the regression error for observation i.

Derivation of Variance of $\hat{\beta}$

$$\begin{aligned}
var[\hat{\beta}|\boldsymbol{X}] &= E[(\hat{\beta}-\beta)(\hat{\beta}-\beta)'|\boldsymbol{X}] \\
&= E[((\beta+(\boldsymbol{X'X})^{-1}\boldsymbol{X'e})-\beta)((\beta+(\boldsymbol{X'X})^{-1}\boldsymbol{X'e})-\beta)'|\boldsymbol{X}] \\
&= E[((\boldsymbol{X'X})^{-1}\boldsymbol{X'e})((\boldsymbol{X'X})^{-1}\boldsymbol{X'e})'|\boldsymbol{X}] \\
&= E[(\boldsymbol{X'X})^{-1}\boldsymbol{X'ee'X}(\boldsymbol{X'X})^{-1}|\boldsymbol{X}] \\
&= (\boldsymbol{X'X})^{-1}\boldsymbol{X'}E[\boldsymbol{ee}'|\boldsymbol{X}]\boldsymbol{X}(\boldsymbol{X'X})^{-1} \\
&= (\boldsymbol{X'X})^{-1}\boldsymbol{X'DX}(\boldsymbol{X'X})^{-1}
\end{aligned}$$

Note that $\boldsymbol{X'DX} = \sum_{i=1}^{n} \mathbf{x}_i\mathbf{x}_i'\sigma_i^2$

The Sandwich Formula in Practice

- The variance $(\boldsymbol{X'X})^{-1}\boldsymbol{X'DX}(\boldsymbol{X'X})^{-1}$ is called the **sandwich formula**: $(\boldsymbol{X'X})^{-1}$ is the "bread" and $\boldsymbol{X'DX}$ is the "meat."
- This is the formula behind every "robust standard error" you see in applied papers.
- In R: `vcovHC(lm_model, type="HC2")` computes this variance using $\hat{e}_i^2$ to estimate the diagonal of $\boldsymbol{D}$.
- Under homoskedasticity ($\boldsymbol{D} = \sigma^2 \boldsymbol{I}$), the sandwich simplifies to $\sigma^2(\boldsymbol{X'X})^{-1}$, which is the classical formula. Robust SEs and classical SEs agree.
- When they diverge, it signals heteroskedasticity in your data.

Special case, $\sigma_i^2 = \sigma_j^2 = \sigma^2$

- If $\boldsymbol{D} = \boldsymbol{I}_n\sigma^2$, (assumption 4*), then we call the error homoskedastic.

$$
\begin{aligned}
var[\hat{\beta}|\boldsymbol{X}] &= (\boldsymbol{X'X})^{-1}\boldsymbol{X'DX}(\boldsymbol{X'X})^{-1} \\
&= (\boldsymbol{X'X})^{-1}\boldsymbol{X'I}_n\sigma^2\boldsymbol{X}(\boldsymbol{X'X})^{-1} \\
&= \sigma^2(\boldsymbol{X'X})^{-1}\boldsymbol{X'X}(\boldsymbol{X'X})^{-1} \\
&= \sigma^2(\boldsymbol{X'X})^{-1}
\end{aligned}
$$

- Homoskedasticity is a convenient assumption, but it also justifies relying on OLS: under $\boldsymbol{D} = \sigma^2\boldsymbol{I}$, Gauss-Markov guarantees no linear estimator can do better.

Gauss-Markov Theorem: Statement

Theorem (Gauss-Markov)

In the homoskedastic linear model (A1–A4*), if $\tilde{\beta}$ is any linear unbiased estimator of β, then

$$\text{var}(\tilde{\beta}|\boldsymbol{X}) \geq \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1} = \text{var}(\hat{\beta}_{OLS}|\boldsymbol{X})$$

The proof has three ideas:

1. What does "linear estimator" mean, and when is one unbiased?
2. Compute the variance of any such estimator.
3. Show the variance difference is always positive semi-definite.

Step 1: What Is a Linear Estimator?

Any estimator that is a linear function of $\boldsymbol{Y}$ can be written as

$$\tilde{\beta} = \boldsymbol{A}'\boldsymbol{Y}$$

for some $n \times k$ matrix $\boldsymbol{A}$ (which may depend on $\boldsymbol{X}$, but not on $\boldsymbol{Y}$).

Examples:

- OLS: $\boldsymbol{A}' = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'$, so $\boldsymbol{A} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}$.
- Sample mean (when $\boldsymbol{X} = \mathbf{1}$): $\boldsymbol{A}' = \frac{1}{n}\mathbf{1}'$.
- Any weighted average of observations is linear in $\boldsymbol{Y}$.

Not linear: median, trimmed mean, $\hat{\beta}/\|\hat{\beta}\|$, any estimator where $\boldsymbol{Y}$ enters nonlinearly.

Step 1 (cont.): When Is $\boldsymbol{A}'\boldsymbol{Y}$ Unbiased?

Substitute $\boldsymbol{Y} = \boldsymbol{X}\beta + \boldsymbol{e}$:

$$\mathbb{E}[\boldsymbol{A}'\boldsymbol{Y}|\boldsymbol{X}] = \boldsymbol{A}'\mathbb{E}[\boldsymbol{Y}|\boldsymbol{X}] = \boldsymbol{A}'\boldsymbol{X}\beta$$

For this to equal β for *every* value of β, we need:

$$\boldsymbol{A}'\boldsymbol{X} = \boldsymbol{I}_k$$

Key constraint: A linear estimator $\boldsymbol{A}'\boldsymbol{Y}$ is unbiased for β if and only if $\boldsymbol{A}'\boldsymbol{X} = \boldsymbol{I}_k$. OLS satisfies this: $(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{X} = \boldsymbol{I}_k$. But many other matrices $\boldsymbol{A}$ do too—the question is which gives the smallest variance.

Step 2: Variance of Any Linear Unbiased Estimator

Since $\tilde{\beta} = \boldsymbol{A}'\boldsymbol{Y} = \boldsymbol{A}'(\boldsymbol{X}\beta + \boldsymbol{e}) = \beta + \boldsymbol{A}'\boldsymbol{e}$:

$$\text{var}(\tilde{\beta}|\boldsymbol{X}) = \boldsymbol{A}' \,\text{var}(\boldsymbol{e}|\boldsymbol{X})\, \boldsymbol{A} = \sigma^2 \boldsymbol{A}'\boldsymbol{A}$$

using homoskedasticity (A4*): $\text{var}(\boldsymbol{e}|\boldsymbol{X}) = \sigma^2 \boldsymbol{I}$.

For OLS specifically, $\boldsymbol{A} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}$, so:

$$\text{var}(\hat{\beta}|\boldsymbol{X}) = \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1} = \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}$$

Goal: Show $\sigma^2\boldsymbol{A}'\boldsymbol{A} \geq \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}$ for any $\boldsymbol{A}$ with $\boldsymbol{A}'\boldsymbol{X} = \boldsymbol{I}_k$.

Step 3: The Decomposition Trick

Define $\boldsymbol{C} = \boldsymbol{A} - \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}$, so $\boldsymbol{A} = \boldsymbol{C} + \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}$.

Why this $\boldsymbol{C}$? It measures the "deviation" of any linear unbiased estimator from OLS. The constraint $\boldsymbol{A}'\boldsymbol{X} = \boldsymbol{I}_k$ forces $\boldsymbol{X}'\boldsymbol{C} = \boldsymbol{X}'\boldsymbol{A} - \boldsymbol{I}_k = \mathbf{0}$.

Now expand $\boldsymbol{A}'\boldsymbol{A}$:

$$
\begin{aligned}
\boldsymbol{A}'\boldsymbol{A} - (\boldsymbol{X}'\boldsymbol{X})^{-1} &= (\boldsymbol{C} + \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1})'(\boldsymbol{C} + \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}) - (\boldsymbol{X}'\boldsymbol{X})^{-1} \\
&= \boldsymbol{C}'\boldsymbol{C} + \underbrace{\boldsymbol{C}'\boldsymbol{X}}_{=\mathbf{0}}(\boldsymbol{X}'\boldsymbol{X})^{-1} + (\boldsymbol{X}'\boldsymbol{X})^{-1}\underbrace{\boldsymbol{X}'\boldsymbol{C}}_{=\mathbf{0}} \\
&\quad + \cancel{(\boldsymbol{X}'\boldsymbol{X})^{-1}} - \cancel{(\boldsymbol{X}'\boldsymbol{X})^{-1}} \\
&= \boldsymbol{C}'\boldsymbol{C}
\end{aligned}
$$

The cross terms vanish because $\boldsymbol{X}'\boldsymbol{C} = \mathbf{0}$—this is why $\boldsymbol{C}$ is defined this way. It produces a *perfect square* on the right-hand side.

Step 3 (cont.): $\boldsymbol{C'C}$ Is Positive Semi-Definite

For any vector $\boldsymbol{v} \in \mathbb{R}^k$:

$$\boldsymbol{v}'\boldsymbol{C}'\boldsymbol{C}\boldsymbol{v} = \|\boldsymbol{C}\boldsymbol{v}\|^2 \geq 0$$

So $\boldsymbol{A}'\boldsymbol{A} - (\boldsymbol{X}'\boldsymbol{X})^{-1} = \boldsymbol{C}'\boldsymbol{C} \geq 0$, and therefore:

$$\text{var}(\tilde{\beta}|\boldsymbol{X}) = \sigma^2\boldsymbol{A}'\boldsymbol{A} \geq \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1} = \text{var}(\hat{\beta}_{OLS}|\boldsymbol{X}) \quad \square$$

No linear unbiased estimator can have lower variance than OLS.
OLS is the **B**est **L**inear **U**nbiased **E**stimator (BLUE).

What Does BLUE Mean for Applied Work?

- OLS is the default starting point *because* under the strongest assumptions no linear unbiased estimator has tighter confidence intervals.
- In practice, homoskedasticity rarely holds exactly, so applied work pairs OLS with **robust standard errors** rather than assuming classical SEs are correct.
- The real value of Gauss-Markov is clarifying what *additional structure* you need to beat OLS:
  - Knowledge of $\Sigma \rightarrow$ GLS can improve efficiency.
  - If $\mathbb{E}[e|\boldsymbol{X}] \neq 0$ (omitted variables, simultaneity), OLS is biased regardless of efficiency $\rightarrow$ need a valid instrument (IV).
  - BLUE says nothing about nonlinear estimators (MLE may dominate if you know the error distribution).

Gauss-Markov in Action: OLS vs. Alternatives

```r
set.seed(42); B <- 5000; n <- 50
b_ols <- b_split <- numeric(B)
for (i in 1:B) {
  x <- rnorm(n); e <- rnorm(n, 0, 2)
  y <- 2 + 3*x + e
  b_ols[i] <- coef(lm(y ~ x))[2]
  h1 <- 1:(n/2); h2 <- (n/2+1):n
  b_split[i] <- (coef(lm(y[h1]~x[h1]))[2]
               + coef(lm(y[h2]~x[h2]))[2])/2
}
c(var(b_ols), var(b_split))
# ~ 0.08        ~ 0.16
```

- DGP: $Y = 2 + 3X + e$, $e \sim N(0, 4)$.
- Split-sample: run OLS on each half, average the slopes. Linear, unbiased—but wastes information.
- OLS variance $\approx$ half the split-sample variance.

Under homoskedasticity, OLS uses all the information in the data. Any other linear unbiased estimator wastes some.

Hansen's Gauss Markov Theorem (Technical)

In the homoskedastic linear model, if $\tilde{\beta}$ is ~~a linear~~ *any* unbiased estimator of β, then

$$\text{var}(\tilde{\beta}|\boldsymbol{X}) \geq \sigma^2(\boldsymbol{X'X})^{-1}$$

This result is derived using the Cramér-Rao bound.

Theoretical Criteria for Estimators

- Gauss Markov: OLS is "best" by reference to any alternative unbiased linear estimator.
- Cramér-Rao bound: the precision (inverse of the variance) of any unbiased estimator is bounded by the Fisher information of the estimator.
- We will define "information" more formally when we get to Maximum Likelihood Estimation.

Method of Moments Estimation

- A **method of moments estimator** (MME) sets sample moments equal to population moments and solves for the parameter.
- Population moment condition: $\mathbb{E}[g(\boldsymbol{x}_i, Y_i, \theta)] = 0$
- Sample analog: $\frac{1}{n}\sum_{i=1}^{n} g(\boldsymbol{x}_i, Y_i, \hat{\theta}) = 0$
- OLS is a method of moments estimator:
  - Population: $\mathbb{E}[\boldsymbol{x}_i(Y_i - \boldsymbol{x}_i'\beta)] = 0 \quad (\text{from } \mathbb{E}[\boldsymbol{x}_i e_i] = 0)$
  - Sample: $\frac{1}{n}\sum_{i=1}^{n} \boldsymbol{x}_i(Y_i - \boldsymbol{x}_i'\hat{\beta}) = 0 \quad \Rightarrow \quad \boldsymbol{X}'(\boldsymbol{Y} - \boldsymbol{X}\hat{\beta}) = 0$
  - Solving gives $\hat{\beta} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{Y}$—the OLS estimator.

Method of Moments for σ^2

- Under homoskedasticity, the population moment is $\mathbb{E}[e_i^2] = \sigma^2$.
- The MME replaces e_i with $\hat{e}_i$ and averages:

$$\hat{\sigma}^2_{MM} = \frac{1}{n}\sum_{i=1}^{n} \hat{e}_i^2$$

- This is biased: $\mathbb{E}[\hat{\sigma}^2_{MM}|\boldsymbol{X}] = \frac{n-k}{n}\sigma^2$ (shown later in the residuals section).
- The bias-corrected version $s^2 = \frac{1}{n-k}\sum \hat{e}_i^2$ is the standard estimator in practice.
- Method of moments generalizes to **GMM** (Generalized Method of Moments) when we have more moment conditions than parameters.

The GLS Assumption

- Switching from OLS to GLS reflects a **substantive claim about the error structure**.
- **Exogenous feature:** We observe (or can estimate) the error covariance Σ—e.g., cross-country data where richer countries have more variable outcomes.
- The claim "I know Σ" is itself a substantive assumption—if wrong, GLS can be *worse* than OLS.
- Two things can go wrong with GLS:
  1. Wrong Σ: you down-weight the wrong observations $\rightarrow$ efficiency loss.
  2. Wrong functional form: if the process is scale-invariant, you should take logs *before* worrying about Σ (recall the invariance slide).

Spherical Errors

- The Gauss Markov Theorem assumed that $E(\boldsymbol{ee'}) = \begin{bmatrix} \sigma^2 & 0 & \cdots & 0 \\ 0 & \sigma^2 & \cdots & 0 \\ \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & \cdots & \sigma^2 \end{bmatrix} = \sigma^2 \boldsymbol{I}$
- In quadratic form, $\sigma^2 \boldsymbol{I}$ is the formula for a sphere

$$\begin{aligned} \boldsymbol{e'}(\sigma^2 \boldsymbol{I})\boldsymbol{e} &= \sigma^2 e_1^2 + \sigma^2 e_2^2 + \cdots + \sigma^2 e_n^2 = q \\ &= e_1^2 + e_2^2 + \cdots + e_n^2 = q/\sigma^2 \end{aligned}$$

Non-Spherical Errors

- More generally we think that errors may exhibit variation: If
$$E(\boldsymbol{ee'}) = \begin{bmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \dots & \vdots \\ 0 & 0 & \cdots & \sigma_N^2 \end{bmatrix}, \sigma_i \neq \sigma_j,$$ then we say we have heteroskedasticity.
- Heteroskedasticity can occur when cases are aggregates of uneven size.
- If $E(\boldsymbol{ee'}) = \begin{bmatrix} \sigma^2 & a & \cdots & b \\ a & \sigma^2 & \cdots & d \\ \vdots & \vdots & \dots & \vdots \\ b & d & \cdots & \sigma^2 \end{bmatrix}, a \neq b \neq d \neq 0$, then we say we have autocorrelation.
- More generally, we will allow for both.

Where Do Non-Spherical Errors Arise?

- **Heteroskedasticity:**
  - Cross-country regressions: richer countries often have more variable outcomes (GDP growth, trade flows).
  - Earnings regressions: variance of wages increases with education and experience.
  - Any regression where the outcome is bounded below (e.g. expenditure ≥ 0): variance shrinks near the bound.
- **Autocorrelation:**
  - Time series: an economic shock in one quarter persists into the next.
  - Panel data: unobserved country or individual traits make errors within a unit correlated.
  - Spatial data: neighboring counties share unobserved shocks (weather, policy spillovers).
- In all these cases, OLS $\hat{\beta}$ is still unbiased, but its variance is *not* $\sigma^2(\boldsymbol{X'X})^{-1}$—classical standard errors are wrong.

Before GLS: Is the Specification Right?

- Heteroskedasticity sometimes signals a **misspecified functional form**, not just a variance problem.
- If GDP growth variance increases with GDP level, the underlying process may be *multiplicative*:

$$P_{t+1} = P_t(1 + g_t) \quad \Rightarrow \quad \Delta \log P_t \approx g_t$$

  A log specification makes effects proportional—and may eliminate the heteroskedasticity.
- The key question: **what transformation leaves the effect unchanged?**
  - *Translation-invariant* mechanism (additive accumulation) $\rightarrow$ levels, $\mathbb{E}[Y \mid X] = X'\beta$
  - *Scale-invariant* mechanism (proportional growth) $\rightarrow$ logs, $\mathbb{E}[\log Y \mid X] = X'\beta$
- In a diff-in-diff design, this distinction matters directly: parallel trends in *levels* vs. parallel trends in *growth rates* are different identifying assumptions.
- GLS addresses heterogeneity that remains *after* the right specification is chosen.

Generalized Least Squares

- Consider the following linear regression model:

$$\boldsymbol{y} = \boldsymbol{X}\beta + \boldsymbol{e}$$

- Allow for heteroskedasticity or correlation in the errors:

$$\mathbb{E}[\boldsymbol{e}|\boldsymbol{X}] = 0$$

$$\text{var}[\boldsymbol{e}|\boldsymbol{X}] = \Sigma\sigma^2$$

- Here Σ is $n \times n$ and may be a function of $\boldsymbol{X}$, σ^2 is the common component to the diagonals (could be 1).
- Recall, Gauss-Markov proves we can do no better than OLS when $\text{var}[\boldsymbol{e}|\boldsymbol{X}] = \sigma^2\boldsymbol{I}$. But now we can do better.

Derivation of Variance (again)

$$
\begin{aligned}
var[\hat{\beta}] &= E[(\hat{\beta}-\beta)(\hat{\beta}-\beta)'] \\
&= E[((\beta+(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{e})-\beta)((\beta+(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{e})-\beta)'] \\
&= E[((\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{e})((\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{e})'] \\
&= E[(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{e}\boldsymbol{e}'\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}] \\
&= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'E[\boldsymbol{e}\boldsymbol{e}']\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1} \\
&= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\Sigma\sigma^2\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1} \\
&= \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\Sigma\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}
\end{aligned}
$$

Lower Bound on Variance of Estimators

Theorem: Given assumptions of linear model, if $\tilde{\beta}$ is an unbiased estimator of β, then

$$\text{var}[\tilde{\beta}] \geq \sigma^2(\boldsymbol{X}'\Sigma^{-1}\boldsymbol{X})^{-1}$$

All we need is to know Σ^{-1}, then correct for it.

Efficiency Lower Bound: Setup (Hansen Ex. 4.6)

Theorem: If $\tilde{\beta} = \boldsymbol{A'Y}$ is linear and unbiased, then $\text{var}(\tilde{\beta}|\boldsymbol{X}) \geq \sigma^2(\boldsymbol{X}'\Sigma^{-1}\boldsymbol{X})^{-1}$.

Proof: Transform to the spherical model via $\Sigma^{-1/2}$:

1. Define $\tilde{\boldsymbol{Y}} = \Sigma^{-1/2}\boldsymbol{Y}$, $\tilde{\boldsymbol{X}} = \Sigma^{-1/2}\boldsymbol{X}$, $\tilde{\boldsymbol{e}} = \Sigma^{-1/2}\boldsymbol{e}$, so $\text{var}[\tilde{\boldsymbol{e}}|\boldsymbol{X}] = \sigma^2\boldsymbol{I}_n$.
2. Write $\tilde{\beta} = \boldsymbol{A'Y} = \boldsymbol{A}'\Sigma^{1/2}\tilde{\boldsymbol{Y}} \equiv \tilde{\boldsymbol{A}}'\tilde{\boldsymbol{Y}}$.
3. Unbiasedness: $\tilde{\boldsymbol{A}}'\tilde{\boldsymbol{X}} = \boldsymbol{A}'\Sigma^{1/2}\Sigma^{-1/2}\boldsymbol{X} = \boldsymbol{A'X} = \boldsymbol{I}_k$.

Efficiency Lower Bound: Applying Gauss-Markov

4. Apply Gauss-Markov to the transformed (spherical) model:

$$\text{var}[\tilde{\beta}|\boldsymbol{X}] = \sigma^2 \tilde{\boldsymbol{A}}' \tilde{\boldsymbol{A}} \geq \sigma^2 (\tilde{\boldsymbol{X}}' \tilde{\boldsymbol{X}})^{-1} = \sigma^2 (\boldsymbol{X}' \Sigma^{-1} \boldsymbol{X})^{-1} \quad \square$$

Intuition: $\Sigma^{-1/2}$ converts any non-spherical model into a spherical one. In the spherical world, OLS (= GLS on original data) is already BLUE.

Aside: Matrix Decomposition

- Any matrix $\mathbf{\Sigma}$ that is positive definite and symmetric can be factored:

$$\mathbf{\Sigma} = \boldsymbol{C}\mathbf{\Lambda}\boldsymbol{C}'$$

- The columns of $\boldsymbol{C}$ are the eigenvectors of $\mathbf{\Sigma}$.
- $\mathbf{\Lambda}$ is a diagonal matrix of the eigenvalues of $\mathbf{\Sigma}$.
- $\mathbf{\Lambda}^{1/2}$ is a diagonal matrix of the square roots of the eigenvalues of $\mathbf{\Sigma}$.

$$\mathbf{\Sigma} = \boldsymbol{C}\mathbf{\Lambda}\boldsymbol{C}' = \boldsymbol{C}\mathbf{\Lambda}^{1/2}\mathbf{\Lambda}^{1/2}\boldsymbol{C}' = \boldsymbol{T}\boldsymbol{T}'$$

$$\mathbf{\Sigma}^{-1} = \boldsymbol{C}\mathbf{\Lambda}^{-1}\boldsymbol{C}' = \boldsymbol{C}\mathbf{\Lambda}^{-1/2}\mathbf{\Lambda}^{-1/2}\boldsymbol{C}' = \mathbf{\Sigma}^{-\mathbf{1/2}}(\mathbf{\Sigma}^{-\mathbf{1/2}})'$$

Aitken (1935) Generalized Least Squares

- If we know Σ, we can do no better than to premultiply our linear model with $\Sigma^{-1/2}$,

$$
\begin{aligned}
\boldsymbol{y} &= \boldsymbol{X}\beta + \boldsymbol{e} \\
\Sigma^{-1/2}\boldsymbol{y} &= \Sigma^{-1/2}\boldsymbol{X}\beta + \Sigma^{-1/2}\boldsymbol{e} \\
\tilde{\boldsymbol{y}} &= \tilde{\boldsymbol{X}}\beta + \tilde{\boldsymbol{e}} \\
\tilde{\beta}_{GLS} &= (\tilde{\boldsymbol{X}}'\tilde{\boldsymbol{X}})^{-1}\tilde{\boldsymbol{X}}'\tilde{\boldsymbol{y}} \\
&= ((\Sigma^{-1/2}\boldsymbol{X})'(\Sigma^{-1/2}\boldsymbol{X}))^{-1}(\Sigma^{-1/2}\boldsymbol{X})'(\Sigma^{-1/2}\boldsymbol{y}) \\
&= (\boldsymbol{X}'\Sigma^{-1}\boldsymbol{X})^{-1}\boldsymbol{X}'\Sigma^{-1}\boldsymbol{y}
\end{aligned}
$$

Here we have $E[\tilde{\beta}_{GLS}] = \beta$ and $var(\tilde{\beta}_{GLS}) = \sigma^2(\boldsymbol{X}'\Sigma^{-1}\boldsymbol{X})^{-1}$

GLS is BLUE

- Suppose b is an alternative linear unbiased estimator that differs by $\boldsymbol{A}$

$$
\begin{aligned}
b &= [(\boldsymbol{X}'\Sigma^{-\mathbf{1}}\boldsymbol{X})^{-1}\boldsymbol{X}'\Sigma^{-1} + \boldsymbol{A}]\boldsymbol{Y} \\
&= [(\boldsymbol{X}'\Sigma^{-\mathbf{1}}\boldsymbol{X})^{-1}\boldsymbol{X}'\Sigma^{-1} + \boldsymbol{A}](\boldsymbol{X}\beta + \boldsymbol{e}) \\
&= [(\boldsymbol{X}'\Sigma^{-\mathbf{1}}\boldsymbol{X})^{-1}\boldsymbol{X}'\Sigma^{-1} + \boldsymbol{A}]\boldsymbol{X}\beta + [(\boldsymbol{X}'\Sigma^{-\mathbf{1}}\boldsymbol{X})^{-1}\boldsymbol{X}'\Sigma^{-1} + \boldsymbol{A}]\boldsymbol{e} \\
\boldsymbol{AX} &= 0 \qquad \text{(Because b is unbiased.)} \\
var(b) &= \sigma^2[(\boldsymbol{X}'\Sigma^{-\mathbf{1}}\boldsymbol{X})^{-1}\boldsymbol{X}'\Sigma^{-1} + \boldsymbol{A}]\Sigma[(\boldsymbol{X}'\Sigma^{-\mathbf{1}}\boldsymbol{X})^{-1}\boldsymbol{X}'\Sigma^{-1} + \boldsymbol{A}]' \\
&= \sigma^2[(\boldsymbol{X}'\Sigma^{-\mathbf{1}}\boldsymbol{X})^{-1} + \boldsymbol{A}\Sigma\boldsymbol{A}' + (\boldsymbol{X}'\Sigma^{-\mathbf{1}}\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{A}' + \boldsymbol{AX}(\boldsymbol{X}'\Sigma^{-\mathbf{1}}\boldsymbol{X})^{-1}] \\
&= \sigma^2[(\boldsymbol{X}'\Sigma^{-\mathbf{1}}\boldsymbol{X})^{-1} + \boldsymbol{A}\Sigma\boldsymbol{A}' + \mathbf{0} + \mathbf{0}] \\
&= \sigma^2(\boldsymbol{X}'\Sigma^{-\mathbf{1}}\boldsymbol{X})^{-1} + \sigma^2\boldsymbol{A}\Sigma\boldsymbol{A}'
\end{aligned}
$$

- Σ is positive definite, so $\sigma^2\boldsymbol{A}\Sigma\boldsymbol{A}' \geq 0$; minimum variance requires $\boldsymbol{A} = 0$ $\square$

GLS Properties

- GLS is unbiased ($\mathbb{E}[\hat{\beta}_{GLS}|\boldsymbol{X}] = \beta$) and BLUE under the correct Σ.
- Algebraically, GLS is OLS on the transformed data $(\tilde{\boldsymbol{y}}, \tilde{\boldsymbol{X}})$. The only structural difference is that the GLS "hat matrix" $\boldsymbol{X}(\boldsymbol{X}'\Sigma^{-1}\boldsymbol{X})^{-1}\boldsymbol{X}'\Sigma^{-1}$ is not symmetric.
- **Feasibility problem:** GLS requires knowing Σ, which is rarely available in practice.
- **Solution:** Feasible GLS (FGLS) estimates Σ in a first step (e.g. from OLS residuals), then applies GLS with $\hat{\Sigma}$. Asymptotically equivalent to GLS under regularity conditions.

When GLS Beats OLS: A Two-Group Example

- Survey data from two groups: Group A (rich, $n_A = 50$, $\sigma_A^2 = 100$) and Group B (poor, $n_B = 50$, $\sigma_B^2 = 1$).
- OLS gives *equal weight* to every observation $\rightarrow$ noisy Group A observations inflate variance.
- GLS (= WLS here) weights by $w_i = 1/\sigma_i^2$: Group A gets weight $1/100$, Group B gets weight 1.

Variance comparison (scalar case, equal group sizes):

$$\text{var}(\hat{\beta}_{OLS}) \propto \frac{\sigma_A^2 + \sigma_B^2}{2} = \frac{101}{2} \qquad \text{var}(\hat{\beta}_{GLS}) \propto \frac{1}{1/\sigma_A^2 + 1/\sigma_B^2} = \frac{100}{101} \approx 1$$

GLS Idea: when you know which observations are noisier, use that information.

Comparing OLS and WLS: R Example

```r
set.seed(99); B <- 5000; n <- 100
b_ols <- b_wls <- numeric(B)

for (i in 1:B) {
  x <- rnorm(n)
  # Two groups: first 50 noisy, last 50 precise
  sigma <- c(rep(10, 50), rep(1, 50))
  y <- 1 + 2*x + rnorm(n, 0, sigma)

  b_ols[i] <- coef(lm(y ~ x))[2]
  b_wls[i] <- coef(lm(y ~ x, weights = 1/sigma^2))[2]
}
c(sd(b_ols), sd(b_wls))   # WLS SE is smaller
# ~ 1.01       ~ 0.14
```

- Both estimators are unbiased ($\bar{\hat{\beta}} \approx 2$), but WLS standard errors are $\sim 7\times$ smaller.
- In practice, use `lm(..., weights = 1/sigma_hat^2)` when group variances are known or estimable.

Lin (2013) Regression Adjustment

- Many of you saw Lin's estimator in your causal inference course. Today we connect it to efficiency.
- **Setup:** Binary treatment $D_i \in \{0,1\}$, outcome Y_i, covariates X_i. Estimand: $\tau = \mathbb{E}[Y(1) - Y(0)]$.
- **Difference in means** (no covariates):

$$\hat{\tau}_{DM} = \bar{Y}_1 - \bar{Y}_0$$

  Unbiased under randomization, but ignores covariate information.
- **Lin's interacted estimator:** regress Y_i on 1, D_i, $(X_i - \bar{X})$, $D_i(X_i - \bar{X})$:

$$Y_i = \alpha + \tau D_i + \gamma(X_i - \bar{X}) + \delta\big[D_i(X_i - \bar{X})\big] + u_i$$

  Report $\hat{\tau}_{\text{Lin}}$ (coefficient on D_i).

Lin Estimator via FWL

- The interaction lets treated and control groups have **separate slopes**.
- By Frisch–Waugh–Lovell, $\hat{\tau}_{\text{Lin}}$ equals the difference in means of **residualized** outcomes:
  1. Within treated: regress Y_i on $(X_i - \bar{X})$, get residuals $\tilde{Y}_i^{(1)}$.
  2. Within controls: regress Y_i on $(X_i - \bar{X})$, get residuals $\tilde{Y}_i^{(0)}$.

$$\hat{\tau}_{\text{Lin}} = \overline{\tilde{Y}}^{(1)} - \overline{\tilde{Y}}^{(0)}$$

- Centering at $\bar{X}$ ensures $\hat{\tau}_{\text{Lin}}$ estimates the ATE (not the effect at some other covariate value).
- You will prove this on Homework 2.

Lin as Variance Reduction: The GLS Analogy

- **Recall the GLS idea:** OLS ignores heteroskedasticity $\rightarrow$ valid but inefficient. GLS uses the error structure to reduce variance.
- **Same logic here:** Difference-in-means ignores covariates $\rightarrow$ valid but leaves explainable variation in the residual.
- Lin's estimator absorbs covariate variation **within each treatment arm**, shrinking the residual variance—analogous to how GLS reweights observations.
- Without the interaction (common slopes across groups), you impose a constraint that can distort the estimate when treatment effects are heterogeneous—just as GLS with a wrong Σ can lose efficiency.

Where the Analogy Breaks Down

- **Source of unbiasedness:**
  - GLS: unbiased because the model is correctly specified ($\boldsymbol{y} = \boldsymbol{X}\beta + \boldsymbol{e}$, $\mathbb{E}[\boldsymbol{e}|\boldsymbol{X}] = 0$).
  - Lin: unbiased because treatment is randomized—even if $\mathbb{E}[Y|X]$ is nonlinear and the regression is "wrong."
- **What gets used:**
  - GLS exploits the $n \times n$ error covariance Σ to *reweight* observations.
  - Lin exploits covariates X to *partial out* predictable variation. It never touches Σ.
- **What misspecification costs you:**
  - Wrong Σ in GLS $\rightarrow$ you lose the BLUE guarantee (efficiency), but $\hat{\beta}$ is still unbiased.
  - Wrong functional form in Lin $\rightarrow$ you still get an unbiased ATE (randomization protects you), but you leave variance on the table.
  - Wrong functional form in OLS *without* randomization $\rightarrow$ bias.
- **Bottom line:** GLS is a *model-based* efficiency gain. Lin is a *design-based* efficiency gain. Randomization does the heavy lifting that correct specification of Σ does for GLS.

Residuals

- We will be using estimates of the residuals to construct covariance matrix estimators that do not require homoskedasticity.
- The residuals $\hat{e}_i = Y_i - \boldsymbol{x}_i'\hat{\beta}$ can be written in vector notation as $\hat{\boldsymbol{e}} = \boldsymbol{Me}$.
- Recall $\boldsymbol{M} = \boldsymbol{I} - \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'$
- Given $\boldsymbol{X}$, the expectation of the residuals is zero :

$$\mathbb{E}[\hat{\boldsymbol{e}}|\boldsymbol{X}] = \mathbb{E}[\boldsymbol{Me}|\boldsymbol{X}] = \boldsymbol{M}\mathbb{E}[\boldsymbol{e}|\boldsymbol{X}] = 0$$

- Given $\boldsymbol{X}$, the variance of the residuals is:

$$\text{var}[\hat{\boldsymbol{e}}|\boldsymbol{X}] = \text{var}[\boldsymbol{Me}|\boldsymbol{X}] = \boldsymbol{M}\text{var}[\boldsymbol{e}|\boldsymbol{X}]\boldsymbol{M} = \boldsymbol{MDM}$$

Homoskedastic Errors

- If we assume homoskedasticity, $\mathbb{E}[e^2|\boldsymbol{X}] = \sigma^2$

$$\text{var}[\hat{\boldsymbol{e}}|\boldsymbol{X}] = \text{var}[\boldsymbol{Me}|\boldsymbol{X}] = \boldsymbol{M}\text{var}[\boldsymbol{e}|\boldsymbol{X}]\boldsymbol{M} = \boldsymbol{MI}\sigma^2\boldsymbol{M} = \boldsymbol{M}\sigma^2$$

- Note that the ith diagonal element of $\boldsymbol{M}$ is $1 - h_{ii}$, so

$$\text{var}[\hat{e}_i|\boldsymbol{X}] = \mathbb{E}[\hat{e}_i^2|\boldsymbol{X}] = (1 - h_{ii})\sigma^2 \neq \sigma^2$$

- $\hat{e}_i^2$ is a biased estimator and it is heteroskedastic.

Prediction Errors

- The prediction errors $\tilde{e}_i = Y_i - \boldsymbol{x}_i \hat{\beta}_{-i} = (1 - h_{ii})^{-1} \hat{e}_i$.
- We defined $\boldsymbol{M}^* = diag\{(1 - h_{11})^{-1},\ (1 - h_{22})^{-1}, \ldots, (1 - h_{nn})^{-1}\}$.
- $\tilde{\boldsymbol{e}} = \boldsymbol{M}^* \hat{\boldsymbol{e}} = \boldsymbol{M}^* \boldsymbol{M} \boldsymbol{e}$
- Given $\boldsymbol{X}$, the expectation of the prediction errors is zero :

$$\mathbb{E}[\tilde{\boldsymbol{e}}|\boldsymbol{X}] = \boldsymbol{M}^* \boldsymbol{M} \mathbb{E}[\boldsymbol{e}|\boldsymbol{X}] = 0$$

- Given $\boldsymbol{X}$, the variance of the prediction errors is:

$$\text{var}[\tilde{\boldsymbol{e}}|\boldsymbol{X}] = \text{var}[\boldsymbol{M}^* \boldsymbol{M} \boldsymbol{e}|\boldsymbol{X}] = \boldsymbol{M}^* \boldsymbol{M} \text{var}[\boldsymbol{e}|\boldsymbol{X}] \boldsymbol{M} \boldsymbol{M}^* = \boldsymbol{M}^* \boldsymbol{M} \boldsymbol{D} \boldsymbol{M} \boldsymbol{M}^*$$

Prediction Errors under homoskedasticity

- Under homoskedasticity, the variance of the prediction errors is:

$$\text{var}[\tilde{\boldsymbol{e}}|\boldsymbol{X}] = \text{var}[\boldsymbol{M}^*\boldsymbol{M}\boldsymbol{e}|\boldsymbol{X}] = \boldsymbol{M}^*\boldsymbol{M}\text{var}[\boldsymbol{e}|\boldsymbol{X}]\boldsymbol{M}\boldsymbol{M}^* = \boldsymbol{M}^*\boldsymbol{M}\sigma^2\boldsymbol{M}\boldsymbol{M}^* = \boldsymbol{M}^*\boldsymbol{M}\boldsymbol{M}^*\sigma^2$$

- The variance of the ith prediction error is then:

$$\begin{aligned}\text{var}[\tilde{\boldsymbol{e}}|\boldsymbol{X}] &= \mathbb{E}[\tilde{e}_i^2|\boldsymbol{X}] \\ &= (1-h_{ii})^{-1}(1-h_{ii})(1-h_{ii})^{-1}\sigma^2 \\ &= (1-h_{ii})^{-1}\sigma^2\end{aligned}$$

- $\sum \tilde{e}_i^2 = \sum((1-h_{ii})^{-1}\hat{e}_i)^2 = PRESS$, "predictive error".

Standardized Residuals

- $\text{var}[\hat{e}_i|\boldsymbol{X}] = (1 - h_{ii})\sigma^2$ varies with $\boldsymbol{X}$
- To make it constant, we can scale the residuals by $(1 - h_{ii})^{-1/2}$

$$\bar{e}_i = (1 - h_{ii})^{-1/2}\hat{e}_i$$

- $\bar{\boldsymbol{e}} = \boldsymbol{M}^{*1/2}\boldsymbol{M}\boldsymbol{e}$
- $\text{var}[\bar{e}_i|\boldsymbol{X}] = \mathbb{E}[\bar{e}_i^2|\boldsymbol{X}] = \sigma^2$
- If the error is homoskedastic, $\bar{e}_i$ has the same bias and variance as the original errors.
- If the error is heteroskedastic, these standardized residuals are not.
- In R, these are recovered by "rstandard(lmmod)": sometimes compared to -2, 2.

Estimating $\hat{\sigma}^2$, s^2, $\bar{\sigma}^2$

- The error variance σ^2 measures the unexplained part of the regression.
- Three estimators:
  1. The method of moments estimator is $\hat{\sigma}^2 = \frac{1}{n}\sum \hat{e}_i^2$ (Biased)
  2. The bias-corrected estimator is $s^2 = \frac{1}{n-k}\sum \hat{e}_i^2$ (sigma in R.)
  3. The standardized estimator $\bar{\sigma}^2 = \frac{1}{n}\sum_{i=1}^n \bar{e}_i^2 = \frac{1}{n}\sum_{i=1}^n (1-h_{ii})^{-1}\hat{e}_i^2$

Estimating $\hat{\sigma}^2$

- Estimator (1) $\hat{\sigma}^2$ takes the average of the squared residuals: $\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}\hat{e}^2$
- The expectation of the estimator uses the following fact:

$$\hat{\sigma}^2 = \frac{1}{n}\boldsymbol{e}'\boldsymbol{M}\boldsymbol{e} = \frac{1}{n}\text{tr}(\boldsymbol{e}'\boldsymbol{M}\boldsymbol{e}) = \frac{1}{n}\text{tr}(\boldsymbol{M}\boldsymbol{e}\boldsymbol{e}')$$

$$\begin{aligned}
\mathbb{E}[\hat{\sigma}^2|\boldsymbol{X}] &= \frac{1}{n}tr(\mathbb{E}[\boldsymbol{M}\boldsymbol{e}\boldsymbol{e}'|\boldsymbol{X}]) \\
&= \frac{1}{n}tr(\boldsymbol{M}\mathbb{E}[\boldsymbol{e}\boldsymbol{e}'|\boldsymbol{X}]) \\
&= \frac{1}{n}tr(\boldsymbol{M}\boldsymbol{D}) \\
&= \frac{1}{n}\sum_{i=1}^{n}(1-h_{ii})\sigma_i^2
\end{aligned}$$

Estimating $\hat{\sigma}^2$

- Under conditional homoskedasticity

$$
\begin{aligned}
\mathbb{E}[\hat{\sigma}^2|\boldsymbol{X}] &= \frac{1}{n} tr(\mathbb{E}[\boldsymbol{Mee'}|\boldsymbol{X}]) \\
&= \frac{1}{n} tr(\boldsymbol{M}\mathbb{E}[\boldsymbol{ee'}|\boldsymbol{X}]) \\
&= \frac{1}{n} tr(\boldsymbol{MI}_n\sigma^2) \\
&= \sigma^2 \frac{n-k}{n}
\end{aligned}
$$

What should I report?

- The bias-corrected error variance estimator s^2 is used throughout applied work.
- It is used to calculate standard errors, F-tests, t-test, and confidence intervals.
- It is typically reported as the RMSE $= \sqrt{s^2}$
- However, s^2 assumes homoskedasticity:
  - You will use robust sandwich estimators like HC2 and HC3 (more on that later),
  - you can report $\bar{\sigma}^2$, which is unbiased under heteroskedasticity.